



## Unit-1.2

# Improper Integrals, Gamma Function & Beta Function

$$\int_{-\infty}^{\infty} f(x) dx$$



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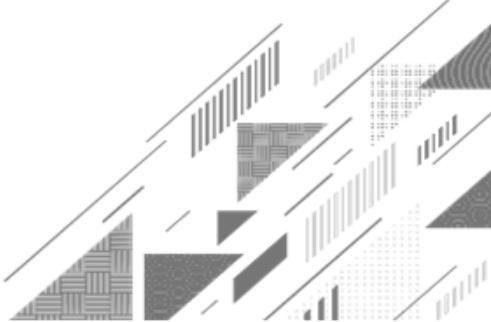
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# OUTLINE

- Method:6 ➡ Improper Integral of First Kind
  - Method:7 ➡ Improper Integral of Second Kind
  - Method:8 ➡ Convergence of Improper Integral of First Kind
  - Method:9 ➡ Convergence of Improper Integral of Second Kind
  - Method:10 ➡ Example on Gamma Function and Beta Function
- 

# Improper Integral

➔ If the **limit** of integral is **infinite** (one or both) or/and **integrand function** is **discontinuous** for some value(s) on the interval of given integral then such integral is known as an improper integral.

## Examples

$$(1). \int_0^{\infty} \frac{1}{x^2 + 1} dx$$

$$(2). \int_{-\infty}^0 x \sin x dx$$

$$(3). \int_{-\infty}^{\infty} e^x dx$$

$$(4). \int_0^1 \frac{1}{x^2} dx$$

$$(5). \int_0^{\frac{\pi}{2}} \sec x dx$$

$$(6). \int_0^3 \frac{1}{x-1} dx$$

# Types of Improper Integral

➔ There are **three types** of improper integrals

(1). Improper integral of **first kind**.

(2). Improper integral of **second kind**.

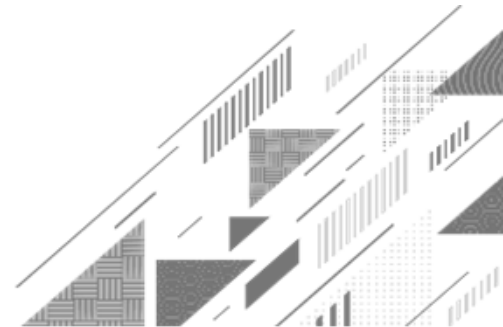
(3). Improper integral of **third kind**(combination of 1<sup>st</sup> & 2<sup>nd</sup> kind).



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## Method:6 ➡ Improper Integral of First Kind



# Improper Integral of First Kind

→ For  $\int_a^b f(x) dx$  either **a** or **b** or **both(a & b)** are **infinite**, then such integral is known as an improper integral of first kind.

## Examples

$$(1). \int_0^{\infty} \frac{dx}{x^2 + 1}$$

$$(2). \int_{-\infty}^0 x \sin x dx$$

$$(3). \int_{-\infty}^{\infty} e^x dx$$

## Sub-types of Improper Integral of First Kind

(1). If  $f(x)$  is continuous on  $[a, \infty)$  then

$$\int_a^{\infty} f(x) \, dx = \lim_{b \rightarrow \infty} \int_a^b f(x) \, dx.$$

(2). If  $f(x)$  is continuous on  $(-\infty, b]$  then

$$\int_{-\infty}^b f(x) \, dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) \, dx.$$

(3). If  $f(x)$  is continuous on  $(-\infty, \infty)$  then

$$\int_{-\infty}^{\infty} f(x) \, dx = \int_{-\infty}^t f(x) \, dx + \int_t^{\infty} f(x) \, dx = \lim_{a \rightarrow -\infty} \int_a^t f(x) \, dx + \lim_{b \rightarrow \infty} \int_t^b f(x) \, dx.$$

# Improper Integral of First Kind

**Example 1**

Evaluate  $\int_0^{\infty} \frac{1}{x^2+1} dx$ .

**Solution**

$$\int_0^{\infty} \frac{1}{x^2+1} dx = \lim_{b \rightarrow \infty} \int_0^b \frac{1}{x^2+1} dx = \lim_{b \rightarrow \infty} [\tan^{-1}(x)]_0^b$$

$$= \lim_{b \rightarrow \infty} [\tan^{-1}(b) - \tan^{-1}(0)]$$

$$= \lim_{b \rightarrow \infty} [\tan^{-1}(b)] - \tan^{-1}(0) = \frac{\pi}{2} - 0 = \frac{\pi}{2}$$

Hence,  $\int_0^{\infty} \frac{1}{x^2+1} dx = \frac{\pi}{2}$



# Improper Integral of First Kind

**Example 5**

Evaluate  $\int_{-\infty}^0 e^{2x} dx$ .

**Solution**

$$\begin{aligned}\int_{-\infty}^0 e^{2x} dx &= \lim_{a \rightarrow -\infty} \int_a^0 e^{2x} dx = \lim_{a \rightarrow -\infty} \left[ \frac{e^{2x}}{2} \right]_a^0 \\ &= \lim_{a \rightarrow -\infty} \left[ \frac{e^0}{2} - \frac{e^{2a}}{2} \right] = \frac{1}{2} - \lim_{a \rightarrow -\infty} \left[ \frac{e^{2a}}{2} \right] \\ &= \frac{1}{2} - 0 = \frac{1}{2}\end{aligned}$$

Hence,

$$\int_{-\infty}^0 e^{2x} dx = \frac{1}{2}$$

# Improper Integral of First Kind

## Example 9

Evaluate  $\int_{-\infty}^{\infty} e^x dx$ .

## Solution

$$\begin{aligned}\int_{-\infty}^{\infty} e^x dx &= \int_{-\infty}^t e^x dx + \int_t^{\infty} e^x dx \\&= \lim_{a \rightarrow -\infty} \int_a^t e^x dx + \lim_{b \rightarrow \infty} \int_t^b e^x dx \\&= \lim_{a \rightarrow -\infty} [e^x]_a^t + \lim_{b \rightarrow \infty} [e^x]_t^b\end{aligned}$$

# Improper Integral of First Kind

Solution Continue

$$\lim_{a \rightarrow -\infty} [e^x]_a^t + \lim_{b \rightarrow \infty} [e^x]_t^b$$

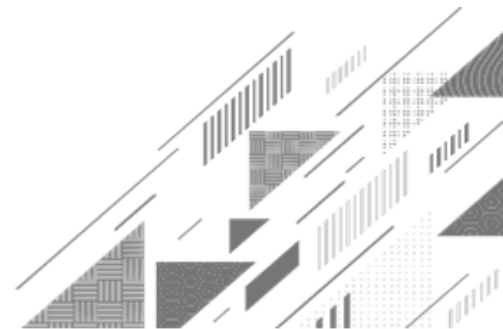
$$= \cancel{e^t} - \lim_{a \rightarrow -\infty} e^a + \lim_{b \rightarrow \infty} e^b - \cancel{e^t}$$

$$= 0 + \infty$$

$$= \infty$$

Hence,  $\int_{-\infty}^{\infty} e^x dx = \infty$

# Method:7 ➡ Improper Integral of Second Kind



# Improper Integral of Second Kind

→ For  $\int_a^b f(x) dx$ ,  $f(x)$  become **discontinuous** at  $x = a$  or  $x = b$  or at **finite** number of points **in** the interval **(a, b)**, then such an integral is known as improper integral of second kind.

## Examples

$$(1). \int_0^1 \frac{1}{x^2} dx$$

$$(2). \int_0^{\frac{\pi}{2}} \sec x dx$$

$$(3). \int_0^3 \frac{1}{x-1} dx$$

## Sub-types of Improper Integral of Second Kind

(1). If  $x = a$  is the point of **discontinuity** for  $f(x)$  then

$$\int_a^b f(x) dx = \lim_{t \rightarrow a} \int_t^b f(x) dx.$$

(2). If  $x = b$  is the point of **discontinuity** for  $f(x)$  then

$$\int_a^b f(x) dx = \lim_{t \rightarrow b} \int_a^t f(x) dx.$$

(3). If  $x = c \in (a, b)$  is the point of **discontinuity** for  $f(x)$  then

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx = \lim_{t \rightarrow c} \int_a^t f(x) dx + \lim_{t \rightarrow c} \int_t^b f(x) dx.$$

# Improper Integral of Second Kind

## Example 9

Evaluate  $\int_0^3 \frac{1}{(x-1)^{\frac{2}{3}}} dx$ .

## Solution

$$\begin{aligned}\int_0^3 \frac{1}{(x-1)^{\frac{2}{3}}} dx &= \int_0^1 \frac{1}{(x-1)^{\frac{2}{3}}} dx + \int_1^3 \frac{1}{(x-1)^{\frac{2}{3}}} dx \\&= \lim_{t \rightarrow 1^-} \int_0^t (x-1)^{-\frac{2}{3}} dx + \lim_{t \rightarrow 1^+} \int_t^3 (x-1)^{-\frac{2}{3}} dx \\&= \lim_{t \rightarrow 1^-} \left[ \frac{(x-1)^{\frac{1}{3}}}{\frac{1}{3}} \right]_0^t + \lim_{t \rightarrow 1^+} \left[ \frac{(x-1)^{\frac{1}{3}}}{\frac{1}{3}} \right]_t^3\end{aligned}$$

## Improper Integral of Second Kind

Solution Continue

$$\begin{aligned} & \lim_{t \rightarrow 1} \left[ 3(x-1)^{\frac{1}{3}} \right]_0^t + \lim_{t \rightarrow 1} \left[ 3(x-1)^{\frac{1}{3}} \right]_t^3 \\ &= \lim_{t \rightarrow 1} \left[ 3(t-1)^{\frac{1}{3}} \right] - 3(-1)^{\frac{1}{3}} + 3(2)^{\frac{1}{3}} - \lim_{t \rightarrow 1} \left[ 3(t-1)^{\frac{1}{3}} \right] \\ &= 0 - 3(-1)^{\frac{1}{3}} + 3(2)^{\frac{1}{3}} - 0 \\ &= 3 \left[ (2)^{\frac{1}{3}} - (-1)^{\frac{1}{3}} \right] \end{aligned}$$

Hence,  $\int_0^3 \frac{1}{(x-1)^{\frac{2}{3}}} dx = 3 \left[ (2)^{\frac{1}{3}} - (-1)^{\frac{1}{3}} \right]$ .



# Convergence or Divergence of Improper Integral

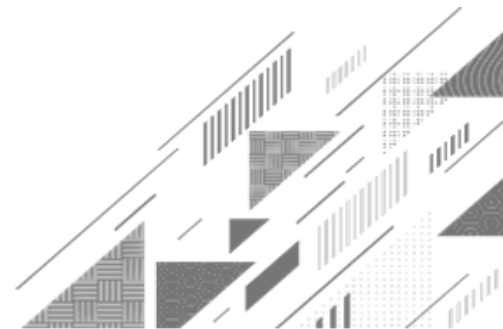
- ➔ If integral has **finite value** then we can say that given integral is **convergent**.
- ➔ If integral has **infinite value** then we can say that given integral is **divergent**.



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# Method:8 → Convergence of Improper Integral of First Kind



# Convergence of Improper Integral of First Kind

**Example 8** Investigate the convergence of  $\int_5^{\infty} \frac{5x}{(1+x^2)^3} dx$ .



**Solution**  $\int_5^{\infty} \frac{5x}{(1+x^2)^3} dx = \frac{5}{2} \int_5^{\infty} (2x)(1+x^2)^{-3} dx$

$$= \frac{5}{2} \lim_{b \rightarrow \infty} \int_5^b (1+x^2)^{-3} (2x) dx = \frac{5}{2} \lim_{b \rightarrow \infty} \left[ \frac{(1+x^2)^{-2}}{-2} \right]_5^b$$

$$\int [f(x)]^n f'(x) dx = \frac{[f(x)]^{n+1}}{n+1} + c$$

# Convergence of Improper Integral of First Kind

Solution Continue

$$\begin{aligned} & -\frac{5}{4} \lim_{b \rightarrow \infty} \left[ \frac{1}{(1+x^2)^2} \right]_5^b \\ &= -\frac{5}{4} \left[ \lim_{b \rightarrow \infty} \frac{1}{(1+b^2)^2} \right] + \frac{5}{4} \left[ \frac{1}{(1+(5)^2)^2} \right] \\ &= 0 + \frac{5}{4} \left[ \frac{1}{(26)^2} \right] = \frac{5}{2704} \text{ Which is finite.} \end{aligned}$$

Hence,  $\int_5^{\infty} \frac{5x}{(1+x^2)^3} dx$  is convergent.

# P-integral Test

→  $\int_a^{\infty} \frac{1}{x^p} dx$  is **convergent** if  $p > 1$  & **divergent** if  $p \leq 1$ .

→  $\int_0^a \frac{1}{x^p} dx$  is **convergent** if  $p < 1$  & **divergent** if  $p \geq 1$ .

## Direct Comparison Test

➔ If  $f(x)$  and  $g(x)$  are continuous on  $[a, \infty)$  and  $0 \leq f(x) \leq g(x)$  for all  $x \geq a$  then

(1). If  $\int_a^{\infty} g(x) \, dx$  is convergent then  $\int_a^{\infty} f(x) \, dx$  is convergent.

(2). If  $\int_a^{\infty} f(x) \, dx$  is divergent then  $\int_a^{\infty} g(x) \, dx$  is divergent.

# Convergence of Improper Integral of First Kind

**Example 5(1)** Check the convergence of  $\int_1^{\infty} \frac{\sin^2 x}{x^2} dx$ .

**Solution**

We know that  $-1 \leq \sin x \leq 1$

$$\Rightarrow 0 \leq \sin^2 x \leq 1$$

$$\Rightarrow 0 \leq \frac{\sin^2 x}{x^2} \leq \frac{1}{x^2}, \forall x \in [1, \infty)$$

## P-integral Test

→  $\int_a^{\infty} \frac{1}{x^p} dx$  is convergent if  $p > 1$  & divergent if  $p \leq 1$ . We know that  $\int_1^{\infty} \frac{1}{x^2} dx$  is **convergent** by **p-integral test**.

# Convergence of Improper Integral of First Kind

Solution Continue

**Direct Comparison Test**

Direct comparison test

$$\int_1^{\infty} \frac{\sin^2 x}{x^2} dx \text{ is convergent.}$$

→ If  $f(x)$  and  $g(x)$  are continuous on  $[a, \infty)$  and  $0 \leq f(x) \leq g(x)$  for all  $x \geq a$  then

(1). If  $\int_a^{\infty} g(x) dx$  is convergent then  $\int_a^{\infty} f(x) dx$  is convergent.



# Limit Comparison Test

→ If  $f(x)$  and  $g(x)$  are positive and continuous on  $[a, \infty)$  and  $L = \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}$  then

(1). If  $0 < L < \infty$  then  $\int_a^\infty f(x) dx$  and  $\int_a^\infty g(x) dx$  both are convergent or divergent together.

(2). If  $L = 0$  and  $\int_a^\infty g(x) dx$  is convergent then  $\int_a^\infty f(x) dx$  is convergent.

(3). If  $L = \infty$  and  $\int_a^\infty g(x) dx$  is divergent then  $\int_a^\infty f(x) dx$  is divergent.

# Convergence of Improper Integral of First Kind

## Example 10

Check the convergence of  $\int_4^{\infty} \frac{3x + 5}{x^4 + 7} dx$ .

## Solution

$$\text{Let } \int_4^{\infty} \frac{3x + 5}{x^4 + 7} dx = \int_4^{\infty} f(x) dx$$

$$\text{Consider } g(x) = \frac{x}{x^4} = \frac{1}{x^3}$$

$$\text{Now, } L = \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{\frac{3x + 5}{x^4 + 7}}{\frac{1}{x^3}} = \lim_{x \rightarrow \infty} \frac{(x^3)(3x + 5)}{x^4 + 7}$$

# Convergence of Improper Integral of First Kind

Solution Continue

$$\lim_{x \rightarrow \infty} \frac{(x^3)(3x + 5)}{x^4 + 7}$$
$$= \lim_{x \rightarrow \infty} \frac{\cancel{x^4} \left( 3 + \frac{5}{x} \right)}{\cancel{x^4} \left( 1 + \frac{7}{x^4} \right)} = \lim_{x \rightarrow \infty} \frac{\left( 3 + \frac{5}{x} \right)}{\left( 1 + \frac{7}{x^4} \right)} = 3$$

$$\Rightarrow 0 < L = 3 < \infty$$

We know that  $\int_4^{\infty} \frac{1}{x^3} dx$  is convergent by p-integral test.

# Convergence of Improper Integral of First Kind

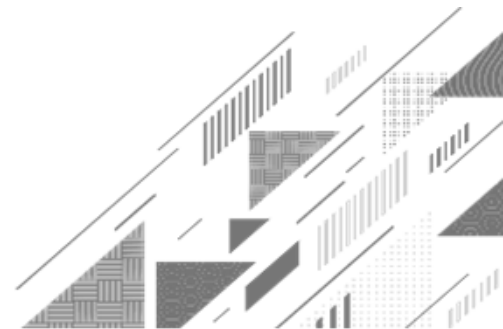
## Solution Continue

Hence, by **limit comparison test**

$$\int_4^{\infty} \frac{3x + 5}{x^4 + 7} dx \text{ is convergent.}$$



## Method:9 ➡ Convergence of Improper Integral of Second Kind



# Convergence of Improper Integral of Second Kind

**Example 5** Define improper integral of both the kinds and check the convergence of

$$\int_0^3 \frac{dx}{\sqrt{9-x^2}}.$$

**Solution**

→ For  $\int_a^b f(x) dx$  either  $a$  or  $b$  or both ( $a$  &  $b$ ) are infinite, then such integral is known as an improper integral of first kind.

# Convergence of Improper Integral of Second Kind

## Solution Continue

→ For  $\int_a^b f(x) dx$ ,  $f(x)$  become discontinuous at  $x = a$  or  $x = b$  or at finite number of points in the interval  $(a, b)$ , then such an integral is known as improper integral of second kind.

$$\int_0^3 \frac{dx}{\sqrt{9-x^2}} = \lim_{b \rightarrow 3} \int_0^b \frac{dx}{\sqrt{9-x^2}} = \lim_{b \rightarrow 3} \left[ \sin^{-1} \left( \frac{x}{3} \right) \right]_0^b$$

# Convergence of Improper Integral of Second Kind

Solution Continue

$$\begin{aligned} & \lim_{b \rightarrow 3} \left[ \sin^{-1} \left( \frac{x}{3} \right) \right]_0^b \\ &= \lim_{b \rightarrow 3} \sin^{-1} \left( \frac{b}{3} \right) - \sin^{-1}(0) \\ &= \sin^{-1}(1) - \sin^{-1}(0) = \frac{\pi}{2} - 0 \\ &= \frac{\pi}{2} \text{ Which is finite.} \end{aligned}$$

Hence,  $\int_0^3 \frac{dx}{\sqrt{9-x^2}}$  is convergent.



# Gamma Function

→ If  $n > 0$ , then **Gamma** function is defined by the integral  $\int_0^{\infty} e^{-x} x^{n-1} dx$   
and it is denoted by  $\Gamma(n)$ . i.e.  $\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx$ .

# Properties of Gamma Function

(1). Reduction formula for Gamma function is  $\Gamma(n + 1) = \Gamma(n)$ ; where  $n > 0$ .

(2). If  $n$  is positive integer, then  $\Gamma(n + 1) = n!$ .

(3). Second form of Gamma function is  $\int_0^{\infty} e^{-x^2} x^{2m-1} dx = \frac{1}{2} \Gamma(m)$ .

(4).  $\Gamma\left(n + \frac{1}{2}\right) = \frac{(2n)! \sqrt{\pi}}{n! 4^n}$  for  $n = 0, 1, 2, 3, \dots$ . For  $n = 0$ ,  $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

$$\text{For } n = 1, \Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}$$

$$\text{For } n = 2, \Gamma\left(\frac{5}{2}\right) = \frac{3\sqrt{\pi}}{4}$$

# Beta Function

→ If  $m > 0, n > 0$ , then **Beta** function is defined by the integral

$$\int_0^1 x^{m-1} (1-x)^{n-1} dx \text{ and it is denoted by } \beta(m, n) \text{ or } B(m, n).$$

$$\text{i.e. } \beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx.$$

# Properties of Beta Function

(1). Beta function is a symmetric function. i.e.  $\beta(m, n) = \beta(n, m)$ , where  $m, n > 0$ .

$$(2). \beta(m, n) = 2 \int_0^{\frac{\pi}{2}} \sin^{2m-1} \theta \cos^{2n-1} \theta \, d\theta.$$

$$(3). \int_0^{\frac{\pi}{2}} \sin^p \theta \cos^q \theta \, d\theta = \frac{1}{2} \beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right).$$

$$(4). \beta(m, n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} \, dx.$$

# Properties of Beta Function

(5). Relation between Beta and Gamma function,  $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ .

## Example on Gamma Function and Beta Function

**Example 2** Define Gamma function and evaluate  $\int_0^{\infty} e^{-x^2} dx$ .

**Solution**

→ If  $n > 0$ , then **Gamma** function is defined by the integral  $\int_0^{\infty} e^{-x} x^{n-1} dx$

and it is denoted by  $\Gamma(n)$ . i.e.  $\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx$ .

## Example on Gamma Function and Beta Function

Solution Continue

$$\int_0^{\infty} e^{-x^2} dx$$

→ Consider  $x^2 = t \Rightarrow x = \sqrt{t}$

$$\Rightarrow 2x dx = dt \Rightarrow dx = \frac{dt}{2x} = \frac{dt}{2\sqrt{t}}$$

→ Now  $x \rightarrow 0 \Rightarrow t \rightarrow 0$  &  $x \rightarrow \infty \Rightarrow t \rightarrow \infty$

$$\Rightarrow \int_0^{\infty} e^{-x^2} dx = \int_0^{\infty} \frac{e^{-t} dt}{2\sqrt{t}} = \frac{1}{2} \int_0^{\infty} e^{-t} t^{-\frac{1}{2}} dt$$

## Example on Gamma Function and Beta Function

Solution Continue

$$\begin{aligned} & \frac{1}{2} \int_0^{\infty} e^{-t} t^{\frac{1}{2}-1} dt \\ &= \frac{1}{2} \Gamma\left(\frac{1}{2}\right) \left( \because \Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx \right) \\ &= \frac{\sqrt{\pi}}{2} \end{aligned}$$

Hence,  $\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$



## Example on Gamma Function and Beta Function

**Example 8** Find  $\beta\left(\frac{7}{2}, \frac{5}{2}\right)$ .

**Solution** We know that  $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ .

$$\begin{aligned}\Rightarrow \beta\left(\frac{7}{2}, \frac{5}{2}\right) &= \frac{\Gamma\left(\frac{7}{2}\right)\Gamma\left(\frac{5}{2}\right)}{\Gamma\left(\frac{7}{2} + \frac{5}{2}\right)} = \frac{\Gamma\left(\frac{7}{2}\right)\Gamma\left(\frac{5}{2}\right)}{\Gamma(6)} \\ &= \frac{\left(\frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi}\right)\left(\frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi}\right)}{(5)!}\end{aligned}$$

## Example on Gamma Function and Beta Function

Solution Continue

$$= \frac{45\pi}{32(120)}$$

$$= \frac{3\pi}{256}$$

Hence,  $\beta\left(\frac{7}{2}, \frac{5}{2}\right) = \frac{3\pi}{256}$

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**THANK  
YOU**

$$\int_{-\infty}^{\infty} f(x) dx$$



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